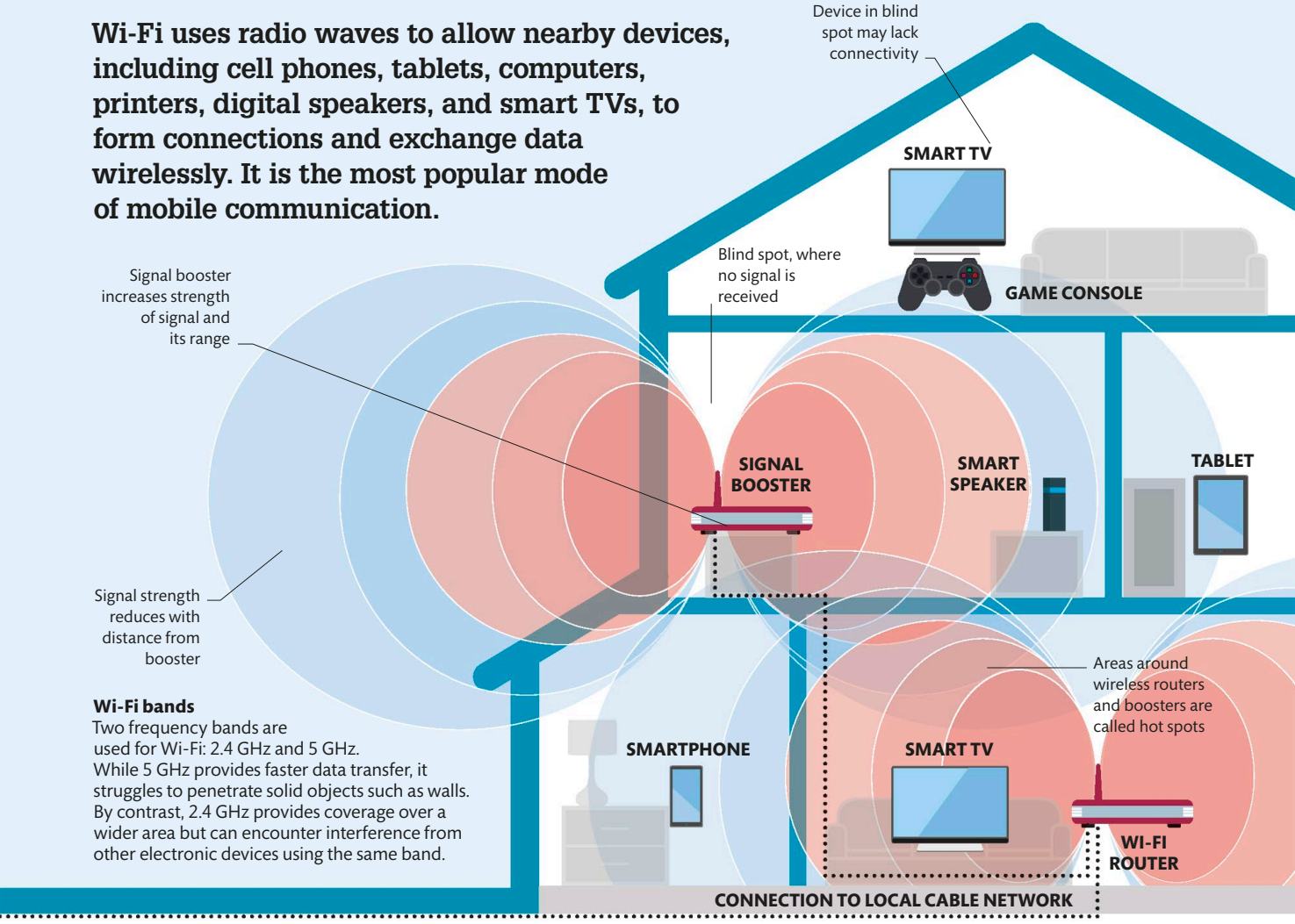
Wi-Fi

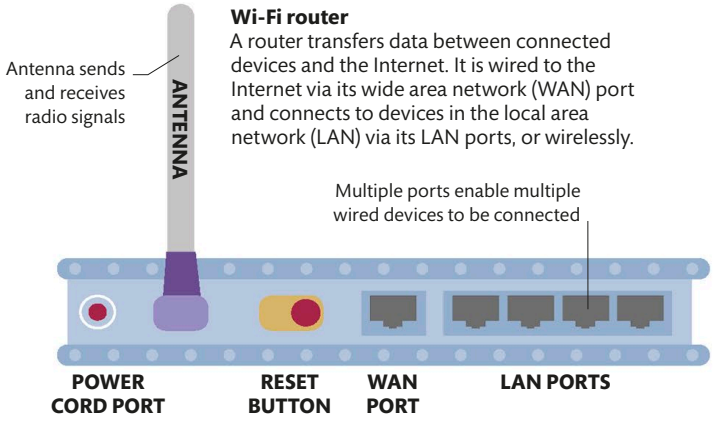
Wi-Fi uses radio waves to allow nearby devices, including cell phones, tablets, computers, printers, digital speakers, and smart TVs, to form connections and exchange data wirelessly. It is the most popular mode of mobile communication.



Wi-Fi bands
Two frequency bands are used for Wi-Fi: 2.4 GHz and 5 GHz. While 5 GHz provides faster data transfer, it struggles to penetrate solid objects such as walls. By contrast, 2.4 GHz provides coverage over a wider area but can encounter interference from other electronic devices using the same band.

How Wi-Fi works

Connecting a device to the Internet using Wi-Fi requires a built-in wireless adaptor—such as an antenna on a cell phone—that transforms digital data into radio signals. When a user sends some form of media, such as a text message or a photo, the adaptor encodes its digital form into radio signals and transmits it to a router. The router then translates the radio signals back into digital data, which it passes to the Internet through a wired connection. This process works the same way in reverse, allowing the wireless exchange of data between devices and the Internet.



Wi-Fi router
A router transfers data between connected devices and the Internet. It is wired to the Internet via its wide area network (WAN) port and connects to devices in the local area network (LAN) via its LAN ports, or wirelessly.